

$P(a_2) \vee Q(a_2)$ has the same truth value as $P(a_2)$. We cannot conclude anything regarding the truth value of $\forall x(P(x) \vee Q(x))$.

If $P(a_2)$ would be False, then $P(a_2) \vee Q(a_2)$ would be False.

If $Q(a_1)$ would be False, then $P(a_1) \vee Q(a_1)$ would be False.

In both cases, by defn. of universal quantifier, $\forall x(P(x) \vee Q(x))$ would be False. But this is not necessarily true. If for all elements a in the domain, at least one of $P(a)$ or $Q(a)$ would be true, then by defn. of OR, $P(a) \vee Q(a)$ would be True and hence $\forall x(P(x) \vee Q(x))$ would be True.

$\therefore$ They are not logically equivalent $\square$

53) Show that $\exists x P(x) \wedge \exists x Q(x)$ and $\exists x (P(x) \wedge Q(x))$ are not logically equivalent.
[Soln:] Let, $\exists x P(x) \wedge \exists x Q(x)$ be True. By defn. of AND, $\exists x P(x)$ is True and $\exists x Q(x)$ is True. By defn. of existential quantifier, there exists at least one element a_1 in the domain s.t. $P(a_1)$ is True. By defn. of existential quantifier, there exists at least one element a_2 in the domain s.t. $Q(a_2)$ is True. By defn. of AND, $P(a_1) \wedge Q(a_1)$ has the same truth value as $Q(a_1)$. By defn. of AND, $P(a_2) \wedge Q(a_2)$ has the same truth value as $P(a_2)$. We cannot conclude anything regarding the truth value of $\exists x (P(x) \wedge Q(x))$.

If $Q(a_1)$ would be True, $P(a_1) \wedge Q(a_1)$ would be True.

If $P(a_2)$ would be True, $P(a_2) \wedge Q(a_2)$ would be True.

In both cases, by defn. of existential quantifier, $\exists x (P(x) \wedge Q(x))$ would be True. But this is not necessarily True. If ^{for} all elements a in the domain, at least one of $P(a)$ or $Q(a)$ would be True, ~~then~~ by defn. of AND, $P(a) \wedge Q(a)$ would be False and by defn. of existential quantifier, $\exists x (P(x) \wedge Q(x))$ would be False.

$\therefore$ They are not logically equivalent $\square$